Does Litigation Risk Matter to Chinese Stock Market?

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Abstract

Our study presents a novel exploration of the impact of litigation risk for the Chinese stock market. Leveraging text-mining techniques to extract 247,000 first-instance litigation records involving listed firms from over 100 million verdicts on China Judgments Online (CJO), we construct a firm-level litigation risk metric. This volume-based measure, calculated as current legal exposure relative to historical baseline, provides a more comprehensive and direct assessment of corporate litigation risk than traditional probit-model estimates of lawsuit probability using securities class action lawsuits (SCA). Empirical results show that firms with high litigation risk usually have lower future stock returns. The litigation risk premium is stronger among micro, high legal fee, and low litigation win rate firms. Moreover, we find that corporate litigation risk deteriorates firm profitability and increases total liabilities.

Highlights

1.We measure the comprehensive litigation risk based on operational litigation cases in China and find its negative relation with expected returns.

2.Litigation risk premium is stronger among micro, high legal fee and low litigation success rate firms.

3.Litigation risk leads to decline in stock prices by impacting the company's profitability and liabilities.

JEL Classification: G12

Keywords: Litigation Risk, Expected Returns

1. Introduction

The pricing power of litigation risk in the Chinese stock market has long been overlooked by both academia and policymakers. Only a limited number of studies have explored the impact of litigation risk on China's economy and society. (e.g., Liu and Zhou 2023; Deng, Xiong & Xiao 2024). And few studies focus on litigation risk of Chinese firms. In contrast, extensive research demonstrates the impact of securities class actions (SCA) on U.S. stock markets and uses SCA lawsuits as a proxy for corporate litigation risk. (e.g., Rogers, Buskirk & Zechman, 2009; Gande and Lewis 2009; Cao et al., 2011; Donelson et al., 2012; Houston et al., 2019; Brogaard et al., 2024). In this study, we measure corporate litigation risk in China based on a variety of lawsuits from China Judgments Online and study how it affects stock prices.

China's large population generates a substantial volume of lawsuits. Since its inception in July 2013, China Judgments Online has collected 120 million legal judgment documents by 2021, including a vast amount of litigation information related to listed companies, thereby greatly enhancing legal transparency. These legal disputes, due to their authenticity and transparency, can reliably reflect the financial distress existing in listed companies. After the disclosure of legal disputes, they may directly impact stock prices through investor sentiment or the reassessment of the company's fundamental value. Consequently, understanding how company litigation risk impacts China's capital market and stock pricing remains a key question.

Why is there so limited literature on the pricing of litigation risk in the Chinese market? The reasons lie in the high difficulty of data collection, the large volume of data, and the significant challenge of extracting useful information from vast amounts of text. Furthermore, unlike in the United States, securities class action (SCA) litigation can directly impact stock prices, which are still rare in China and have not been widely utilized[1]. Scholars study the U.S. benefit from structured and detailed SCA data through resources like the ISS-SCAS database (Brogaard et al., 2024), while comparable

[1] In November 2021, China's first securities class action litigation was adjudicated. As of November 2024, the number of securities class action cases in China remains below 100.

datasets are unavailable in China, making litigation data difficult to obtain.

In this paper, we use web crawler technology to overcome the limitations of collecting litigation data. We collect 120 million legal documents from China Judgments Online for the period from 2010 to 2021. Then we filter out 246,685 trials of first instance related to corporate operations, and extract relevant information such as cause of action, trial procedure, case legal fee, plaintiff fee, defendant fee, and judgement date.

Based on obtained corporate lawsuits, we construct a new indicator to measure the monthly litigation risk for each company, which is calculated by the number of lawsuits in the current month divided by the number of lawsuits in the past three months. Our new indicator incorporates richer information of ex-post legal judgements and variations in litigation risk across the time series for each firm, compared with previous studies that employ the estimated probability of getting sued as the proxy for litigation risk (e.g., Johnson et al. 2001; Rogers and Stocken 2005; Brown et al. 2005).

Using this new measurement in portfolio analysis, we find stocks with high litigation risk generate a substantial monthly premium of -0.33% (t = -2.22), and the negative relation is particularly pronounced among micro, high legal fee and low litigation win rate firms. Based on these findings, our analysis yields two unique effects of legal texts on stock price, compared to traditional media texts and corporate announcements. First, only high-value legal judgments exert a significant negative impact on stock price, whereas low-value judgements do not. Second, investors do not blindly undervalue company's stock price simply due to a higher number of lawsuits it involved, but rather rationally assess the company's litigation outcomes, and then make investment decisions.

We further investigate the economic channels of litigation risk in predicting stock returns by examining the causes of legal judgements involving listed companies. We find sales contract, labor, and construction contract disputes constitute the main components of first-instance cases, accounting for 40.2%, 12.5%, and 10.4%. All of these legal disputes will undermine companies’ profitability and increase their total liabilities, and generate negative abnormal returns. From profitability perspective, lawsuits will generate substantial expenses for company, including legal fee, settlements and attorneys' fees, and some intangible operational impairments which includes increasing the market participants' perception of company uncertainty, investors' expected losses, the erosion of customer bases and supply chain partnerships, and reputational impairment to the company (e.g., Engelmann and Cornell, 1988; Karpoff and Lott, 1993; Karpoff et al., 2008; Gande and Lewis, 2009). From total liabilities perspective, company's involvement in litigation increases its total liability level. Sales contract and construction disputes may delay or prevent the company from collecting receivables from customers, resulting in delayed payments to suppliers (Malm and Sah, 2019) and increasing company's accounts payable. Labor contract disputes will increase employee salaries payable. Other types of cases, such as product liability and property compensation disputes, can also generate significant debt for companies.

Our work contributes to three strands of the literature. First, our paper contributes to the empirical literature measuring corporate litigation risk in non-U.S. country. It is particularly valuable because securities class action (SCA) litigation remains underdeveloped in most countries due to divergent corporate legal systems. Thus, corporate litigation risk measurement has been largely limited to U.S. contexts in existing research. Few scholars have attempted to measure corporate litigation risk across different countries, but face difficulties in data acquisition (e.g., Arena and Ferris, 2018).

The second strand of the literature to which this paper makes a contribution is discovering the pricing power of comprehensive litigation risk in the Chinese market. Our sample consists of a variety of lawsuits, contrasting with existing research on litigation risk only focuses on a single lawsuit, such as SCA litigation (e.g., Rogers, Buskirk & Zechman, 2009; Cao et al., 2011; Donelson et al., 2012; Houston et al., 2019; Brogaard et al., 2024) and patent infringement litigation (e.g., Deng, Xiong & Xiao 2024). This enables the pricing power of litigation risk can be explained by the types of cases it involved.

Finally, our paper also makes contribution in concerning on the case-level information including legal fee and litigation outcomes (whether the company wins the lawsuit). Our advances beyond prior literature that only focused on litigation incidence (whether sued), lawsuit probabilities, and aggregate case volumes (e.g., Johnson et al., 2001; Rogers and Stocken, 2005; Brown et al., 2005; Arena and Ferris, 2018; Deng, Xiong & Xiao, 2024; Brogaard et al., 2024), which fail to distinguish which case truly affect firms. Our paper provides new evidence of combing the contents of law and economics in the view of big data, and has identified the unique pricing power of litigation texts.

The remainder of this paper is recognized as follows. Section 2 describes the data and the litigation risk measure. Section 3 presents the portfolios analysis. Section 4 provides two economic impact channels. Section 5 concludes the paper.

2. Data and Summary Statistics

2.1 Data Description

We obtained 120 million documents from the China Judgments Online[2]. After screening, 4.12 million lawsuits are related to listed companies. We extract the plaintiff and defendant information, cause of action, trial procedure, case legal fee, plaintiff fee, defendant fee, and judgement date from the Judgement on China Judgments Online (Figure 1 provides a judgement sample).

[Insert Figure 1 about here]

After extracting, we match the names of the plaintiff and defendant companies in each document with the names of Chinese listed companies and their subsidiaries, ultimately filtering out the legal document data related to each listed company. From Figure 2, we can see the total number of lawsuits involving listed companies exhibit a

[2] The China Judgment Online was officially put into operation on July 1, 2013. According to the Interim Measures of the Supreme People's Court for the Online Publication of Judgment Documents, except for special circumstances provided by law, the legally effective judgments, rulings and decisions of the Supreme People's Court should generally be published on the Internet. However, by 2023, the local courts have required that the judgment documents are not open and not online, and since 2022, the annual increment of legal documents has decreased significantly, with 28.13 million increasing in 2020, 28.05 million in 2021, and only 10 million in 2022. In 2024, the Supreme Court announced that the "National Court Judgment Library" will be online in January 2024, which regarded as a substitute for the China Judgment Online. The National Court Judgment Library is only available for court personnel to search judgment documents on the internal private network, and is not accessible to lawyers, legal researchers, and the wider public.

consistent year-on-year increase, reaching a peak in 2020 then experiencing a modest downturn in 2021. From Figure 3, we find vehicle traffic accident liability dispute constitutes the majority of lawsuits involving listed companies, followed by financial loan contract disputes and sales contract disputes.

[Insert Figure 2 about here]

[Insert Figure 3 about here]

To screen for lawsuits solely associated with business operations, we exclude vehicle traffic accident liability disputes and other cases unrelated to company activities. Furthermore, we exclude cases involving financial firms, non-first-instance cases, and cases where the listed company are the plaintiff. After these filters, we finally obtain 246,685 first-instance judgments where listed companies are defendants for subsequent analysis.

2.2 Measuring litigation risk

The academic study of litigation risk (the risk of securities class action lawsuits) measurement has a rich history. The vast majority of scholars employ probit model to investigate which characteristics make firms more susceptible to being sued where dependent variable is a binary variable (0 or 1) indicating whether the company got sued, and define litigation risk as the estimated probability of getting sued by probit model (e.g., Johnson et al. 2001; Rogers and Stocken 2005; Brown et al. 2005). Some scholars also explore new measurement for litigation risk, such as directors and officers liability insurance premium (Cao and Narayanamoorthy, 2011) and federal judges' ideology (Brogaard et al., 2024).

Unlike previous studies on litigation risk, our paper mainly concerns how publicly disclosed litigation affects post-judgement stock prices of firms, as opposed to investigating the determinants of probability of company being sued. Accordingly, the litigation risk indicator we develop incorporates both the frequency of occurred litigations and their time-varying intensity patterns.:

$$LR_{i,t} = \frac{Defendant_{i,t}}{\sum_{t-1}^{t-3} Defendant_{i,t}} \tag{1}$$

where $Defendant_{i,t}$ refers to the number of first-instance cases in which company i is involved as a defendant in month t, we use the ratio of the number of cases in the current month to the total number of cases during the past three months as a measure of the litigation risk for the listed company[3].

2.3 Summary Statistics

Our sample includes monthly data from 2010 to 2021 for 2,870 companies listed on the Shanghai Stock Exchange, Shenzhen Stock Exchange, and the Growth Enterprise Market. The financial data of listed companies, information on listed companies' subsidiaries and Fama-French's (2015) five factors are sourced from the China Stock Market and Accounting Research (CSMAR) database. Chinese risk-free rate and Fama-French's (1993) three factors from the RESSET Financial Research Database (RESSET/DB). To avoid the impact of outliers, all continuous variables are winsorized at the 1% level.

[Insert Table 1 about here]

Table 1 shows the descriptive statistics for the main variables, where $Defendant$ refers to the number of first-instance cases where the listed company is the defendant, with only 60% of companies being sued in the first instance monthly on average.

Furthermore, our paper selects several control variables: $Size$ represents firm size, measured by the natural logarithm of total assets; BM represents the book-to-market ratio, calculated as the book value divided by the market value; Age represents firm age, calculated as the current year minus the IPO year; $Growth$ refers to the company's growth rate, measured by the growth rate of the company's operating revenue; LEV represents the degree of leverage, measured by the leverage ratio; IDR represents the board composition, measured by the proportion of independent directors on board; SOE represents the ownership nature of the enterprise, with a value of 1 indicating state-

[3] The measurement of company's litigation risk for a given month should not be limited solely to the number of lawsuits filed during that month. Therefore, when constructing the litigation risk indicator, we considered a longer time span and depicted the company's relative litigation risk each month in the form of a ratio. We also constructed indicators based on a longer time horizon (6 months, 9 months, 12 months), but none performed as well as those based on past 3 months.

owned enterprise and 0 indicating private enterprise. $Profitability$ represents firm profitability, calculated by the operating revenue minus cost of goods sold divided by average of total assets. $In(Liabilities)$ represents firm total liabilities, measured by the natural logarithm of company's total liabilities.

3. Portfolio Analysis

3.1 Univariate analysis of litigation risk

After measuring the monthly litigation risk of Chinese listed companies, we aim to explore whether this litigation risk can affect stock prices. We exclude the stocks that never faced litigation risk[4]. Then we form three monthly rebalanced portfolios (Low, Medium, High) by sorting on firm's lagged litigation risk[5] and then look at their returns in the next month.

[Insert Table 2 about here]

Table 2 shows that litigation risk can predict cross-sectional stock returns. Litigation risk significantly and negatively predicts long-short spread portfolio returns. The long-short spread portfolio generates an average return of -0.33% (t = -2.22) per month, with returns of 0.90% for low litigation risk portfolio versus 0.57% for high litigation portfolio. Since the low, medium and high groups all have litigation risk either in the current month or in the past, they do not outperform the market portfolios (CAPM, FF3 and FF5).

It is worth noting that this effect is not significant in value-weighted portfolios where we present in Table A1. There is little difference in average monthly returns between the low litigation risk group and the high litigation risk group, from 0.40 % for low litigation risk portfolio to 0.38% for high litigation portfolio in value-weighted

[4] We initially intended to include companies that had never faced litigation risk in portfolio analysis, but after grouping them, their performance was poor, showing no significant difference in returns compared to other litigation risk portfolios. We believe that in this "clean" portfolio, companies have never faced litigation risk does not mean they have a higher premium. Without the external condition of litigation risk, their returns are determined by more complex factors, which could be macroeconomic, microeconomic, or other difficult-to-observe variables..

[5] We classify stocks where the numerator in equation (1) is 0 but the denominator is not as low litigation risk. These stocks were sued in the past but not in the current month. Stocks with a result greater than 0 in equation (1) are classified as medium litigation risk. We classify stocks where the numerator is not 0 but the denominator is 0 as high litigation risk, as these stocks have never faced litigation risk before, so being sued in the current month could attract more attention in the short term.

portfolios. The significant effect in equal-weighted portfolios versus the insignificant results in value-weighted portfolios may stem from heterogeneous firm-size responses to litigation risk. Small firms appear more sensitive to litigation risk, while large firms demonstrate greater resilience. This issue will be discussed further in Section 3.2.

3.2 Bivariate analysis of litigation risk

Based on the results from section 3.1, we find the cumulative returns of portfolios constructed based on litigation risk only significant in the equal-weighted calculation, but not in value-weighted calculation. To investigate the reasons behind this, we independently sort firms into three size groups (micro, small, and large) and three litigation risk portfolios in each month, and then look at their returns and alphas in the next month within each size group.

[Insert Table 3 about here]

Table 3 suggests that the negative premium effect of litigation risk is only significant in micro companies, while it is not significant and even reversed in large companies. The long-short spread portfolio generates an average monthly return of -0.51% (t = -2.40) for micro firms. Micro companies have a strong negative risk premium due to litigation risk, but large companies generally do not. As a result, in the value-weighted calculation, the negative returns of micro companies are offset by the positive returns of large companies, leading to insignificant value-weighted results in Table A1.

3.3 Control for legal fee

As mentioned in the introduction, few articles have studied the impact of litigation and its related variables on stock prices. Beck and Bhagat (1997) find the settlement amount of litigation is positive related to the seriousness of allegation in the suit by studying the performance of sued and non-sued firm in shareholder class action litigation. The settlement amount is similar to but different from the total legal fee. Legal fee refers to the end of every legal trial, whether it is the first instance, the second instance or the retrial case, the defendant and plaintiff need to pay to the people's court, if the people's court judges that the defendant is fully responsible, the legal fee are fully borne by the defendant, if two parties are mutually responsible, they will bear legal fee in proportion. Moreover, the higher amount involved in the case itself or amount claimed in the lawsuit, the higher legal fee will be[6].

Settlement amounts and legal fee can both represent the severity of the case. Whether viewed from the perspectives of economic loss, reputational damage, or environmental pollution, larger legal fee or settlement amount reflects a greater impact of the case on the economy and society. Therefore, we posit that higher legal fee associated with a case have a more significant impact on stock price. Based on it, we investigate litigation risk effect among different legal fee groups.

[Insert Table 4 about here]

We aggregate monthly legal fee at firm level, constructing three key monthly frequency variables: total legal fee, plaintiff fee, and defendant fee. Table 4 shows the descriptive statistics for each legal fee group, it can be seen that defendant fee is significantly greater than the plaintiff fee. Listed companies as defendants, are legally judged to bear a greater degree of fault and are required to pay higher legal fee. Based on total legal fee, we independently sort firms into three fee groups (low, medium, and high) and three litigation risk portfolios in each month, and then look at their returns and alphas in the next month within each fee group.

[Insert Table 5 about here]

Table 5 suggests the litigation effect is only significant in high legal fee companies, the long-short spread portfolio generates an average monthly return of -0.68% (t = -2.91) for high legal fee firms, whereas the spread portfolios for other groups show insignificant results. In terms of the results, only cases with high legal fee, which reflect severe impacts on the economy and society, will truly affect a company's stock price. Cases with medium or low legal fee typically do not have a significant impact on

[6] According to the Regulation on the Payment of Legal Fee implemented by the People's Courts since April 1, 2007, the higher amount or value of the claim in a case, the higher legal fee will be. For example, in property cases, the filing fee is 50 yuan for cases with a claim of less than 100,000 yuan, 2.5% for claims between 100,000 yuan and 1,000,000 yuan, 2% for claims between 1,000,000 yuan and 20,000,000 yuan, 1.5% for claims between 20,000,000 yuan and 50,000,000 yuan, and so on. Other cases also generally follow this pattern.

company's stock price.

To further investigate the relationship between legal fee and stock price, we divide the legal fee into plaintiff fee and defendant fee, and analyze their effects on stock prices, separately. The construction of the cumulative portfolio is consistent with the total legal fee.

[Insert Table 6 about here]

Table 6 displays the results for portfolios double sorted on defendant fee and litigation risk. Similar to the pattern in Table 5, the long-short spread portfolio generates a significant monthly return of -0.56% (t = -2.36) only for the high defendant fee firms. And the long-short spread portfolio return remains insignificant in the plaintiff group even for the high plaintiff fee group where we present in Table A2. Therefore, we conclude the factor that exacerbates the litigation risk premium is the portion of the case amount borne by the defendant, rather than that borne by the plaintiff. The larger legal fee involving the company as the defendant, the more severe the impact on the economy and society, and the greater the decline in the company's stock price.

3.4 Considering win rate of lawsuit

In addition to the total legal fee involved in the case amplifying the effects of litigation risk, whether the company, as the defendant, wins each lawsuit should also create heterogeneous impacts on company's stock price in the period following the announcement of judgment. We determine whether the defendant wins the lawsuit by comparing the plaintiff and defendant fee in each case, specifically, if the court orders the defendant to pay higher fee than the plaintiff in the judgment, it indicates the defendant lost the case, according to Liu and Zhou (2023). Next, we aggregate and summarize the win-loss results of all lawsuits where the company is the defendant each month, eventually obtain the company's monthly litigation win rate through equivalent averaging. After that, we independently sort firms into two win rate groups (win rate of 50% or less and win rate of more than 50%) and three litigation risk portfolios in each month, and then look at their returns and alphas in the next month within each win rate group.

[Insert Table 7 about here]

Table 7 shows that whether a company wins a lawsuit indeed affects the magnitude of the litigation risk premium, the long-short spread portfolio generates an average monthly return of -0.32% (t = -2.15) for low win rate group whereas -0.24% (t = -1.29) for high win rate group. We used to perceive that as long as a company involved as defendant in litigation, after the publication of this judgment, it will have a negative impact regardless of whether it ultimately wins or not, leading to a decline in its stock price, similar to negative news or poor earnings reports. However, the results from Table 7 reveal that litigation outcomes differ from negative news or poor earnings reports. If a company's win rate exceeds 50% in a given month, meaning the company wins the majority of its lawsuits during that month, it will not have a significant impact on its stock price though it is classified as high litigation risk firms which measured by the number of lawsuits.

In summary, the disclosure of legal judgments against listed companies as defendants cannot be simply regarded as negative events. And the effects of legal judgments on stock price also differ from traditional news reports and company announcements. Firstly, only high-value legal judgments significantly impact future stock prices, while low-value cases have minimal effects. Secondly, even if a company loses a lawsuit, its share price may remain unaffected, the emergence of negative price shock depends on the company's overall win rate in recent legal judgements. It means investors do not blindly undervalue company's stock price simply due to higher number of lawsuits involved, but rather rationally read judgements outcomes, and then make investment decisions.

Compared to news and company announcements, legal judgments also possess a unique characteristic: they are inherently truthful. News can be fabricated, speculative, or even distort facts (Carvalho, Klagge and Moench, 2011), and company announcements may be falsified (Hoberg and Lewis, 2017) or conceal information (Bertomeu et al., 2022), In contrast, legal judgments are impartial, rigorous, and fully grounded in reality, backed by the authority of national judicial bodies, ensuring their authenticity. Our findings in this section provide empirical evidence for the market pricing of legal judgements, revealing different patterns compared to traditional textual sources like news reports, company disclosures, and policy texts in asset pricing.

4. Economic Explanations

Our paper measures the litigation risk of listed companies in China based on a comprehensive variety of litigation types which differs from America based only on SCA litigation (e.g., Romano, 1991; Gande and Lewis, 2009; Bauer and Braun, 2010; Duanmu et al., 2022). Therefore, the impact of litigation risk on Chinese listed companies should be related to the types of cases that constitute litigation risk. In this section, we first analyze the causes of 246,685 lawsuits where listed companies are the defendants to explore which types of lawsuits dominate and how these lawsuits impact the stock prices.

[Insert Figure 4 about here]

Figure 4 shows that sales contract, labor, and construction contract disputes constitute the main components of first-instance cases where listed company are defendants, accounting for 40.2%, 12.5%, and 10.4%, respectively[7]. These three types of disputes account for approximately 63% of all legal cases. Specifically, Sales contract disputes in listed companies can directly impact their main business and undermine their profitability. First, they may lead to delays in recovering receivables. Second, if the company loses the case, it has to pay additional compensation, these will directly reduce the company's cash flow and constrain business expansion capacities. Labor disputes also significantly impact company's payments for employees, whether it involves paying employee salaries or compensation paid to employees for labor contract violations. Construction contract disputes will involve suppliers' material delivery. These disputes influence the company's profitability and total liabilities.

[7] Due to the lack of a unified standard for contract naming across various courts, there are a total of 721 contract names among the 246,685 contracts. Due to large number of contract names, we only display the top 20 lawsuit causes. Additionally, we aggregate contracts related to sales, such as sales contract, commercial housing contract, product seller liability disputes and so on. The aggregation method for the other two categories of contracts is consistent with this approach.

4.1 Profitability Channel

Companies facing legal lawsuits will generate substantial direct and indirect expenses, these expenses will significantly impair the company's profitability (Arena, 2018) and depress share prices. Direct expenses encompass legal fee, settlements, attorneys' fees, and similar charges, which adversely affect the company's cash flow. And indirect expenses impose far more negative impacts on company's profitability than direct litigation expenses, for instance, increasing the market participants' perception of company uncertainty and expected losses, the erosion of customer bases and supply chain partnerships, and reputational impairment to the company (e.g., Engelmann and Cornell, 1988; Karpoff and Lott, 1993; Karpoff et al., 2008; Gande and Lewis, 2009). As such, our research holds that the negative predictability of litigation risk stems from the damage of lawsuits to company's profitability.

Then, we empirically test the mechanism of company's profitability. According to Jiang, Qi and Tang (2018), we construct profitability indicator[8] for each firm. We use the following regression equation to study the relationship between firm's profitability and its litigation risk.

$$Profitability_{i,t} = \beta_0 + \beta_1 LR_{i,t-1} + \beta_2 Controls_{i,t-1} + \mu_t + \gamma_i + \epsilon_{i,t} \qquad (2)$$

where $Profitability_{i,t}$ refers to the profitability of company i in month t, $LR_{i,t-1}$ refers to the litigation risk of company i in month $t-1$ which is measured by equation (1), $Controls_{i,t-1}$ refers to the control variables listed in Table 1, μ_t refers to the time fixed effect, γ_i refers to the entity fixed effect.

[Insert Table 8 about here]

Table 8 presents the average slopes and their *t*-statistics for the regression of profitability. As shown in column (1) and (2), litigation risk has a negative coefficient of -0.004%, with a *t*-statistics of -2.58 and -2.86, indicating a strong negative

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8 The profitability indicator is computed as follows:

$$GP_t = \frac{OR_t - COGS_t}{0.5(AT_{t-1} + AT_t)}$$

where OR_t is the operating revenue in quarter t, $COGS_t$ is the cost of goods sold in quarter t, AT_t is the average of total assets in quarter t.

predictability for company profitability in the next month. To verify the robustness of the results, we replace independent variable with $Defendant_{i,t-1}$, where number of first-instance cases which company i is defendant. The results in column (3) and (4) remain negative and significant. We also provide results with the explanatory variables lagged by 6 and 12 months in Table A3, ensuring consistency in prediction frequency with semi-annual and annual reports. The results still pronounced. Overall, companies with higher litigation risks tend to have lower profitability and are more likely to be trapped into financial distress, which leads to a decline in the company's stock price.

4.2 Total Liabilities' Channel

Company's involvement in litigation increases its total liability level. Based on Figure 4, sales contract and construction disputes may delay or prevent the company from collecting receivables from customers, resulting in delayed payments to suppliers (Malm and Sah, 2019) and increasing company's accounts payable. Labor contract disputes will put pressure on the company to pay a large amount of employee wages and be required to pay other work-related injury expenses. All these factors exacerbate corporate total liability level, generating negative impact on stock prices.

We obtain liabilities from the balance sheet of listed companies to explore the impact of litigation risk on it. We exploit the following regression equation to study the relationship between firm's liabilities and its litigation risk.

$$In(Liabilities)_{i,t} = \beta_0 + \beta_1 LR_{i,t-1} + \beta_2 Controls_{i,t-1} + \mu_t + \gamma_i + \epsilon_{i,t} \quad (3)$$

where $In(Liabilities)_{i,t}$ refers to the natural logarithm of total liabilities of company i in month t, $LR_{i,t-1}$ refers to the litigation risk of company i in month $t-1$ which is measured by equation (1), $Controls_{i,t-1}$ refers to the control variables listed in Table 1, μ_t refers to the time fixed effect, γ_i refers to the entity fixed effect.

[Insert Table 9 about here]

Table 9 presents the average slopes and their *t*-statistics for the regression of liabilities. As shown in column (1) and (2), litigation risk has a positive coefficient of 0.41% and 0.39%, with a *t*-statistics of 7.81 and 7.50, indicating a strong positive predictability for company liabilities in the next month. We replace independent variable with $Defendant_{i,t-1}$ for robustness test. The results in column (3) and (4) remain positive and significant. We also provide results with the explanatory variables lagged by 6 and 12 months in Table A4. The results still pronounced. This finding suggests that litigation risk will increase company's liability level, which may increase the possibility of forgoing positive NPV projects in the future, and have a negative effect on stock prices (Myers, 1984).

5. Conclusion

Researches on litigation risk in the Chinese market has long been overlooked, particularly in asset pricing domain. To address this gap, our study proposes a comprehensive indicator as the proxy of litigation risk, which differs from existing literature that employ the predicted probability of company being sued as litigation risk. Our paper reveals the negative relationship between litigation risk and expected returns. The litigation risk premium is stronger among micro, high legal fee and low litigation win rate firms. We also find unique pricing implications of legal texts, its negative predictability depends on total case value and company's overall win rate in recent legal judgements, which is different from traditional texts such as media texts and company disclosure texts. We furtherly find that sales contract, labor, and construction contract disputes constitute the defendant lawsuits, and infer the negative predictability of litigation risk is driven by two economic channels: firms' profitability and liabilities.

Our research makes contribution for measuring corporate litigation risk in non-U.S. country, future researchers can obtain corporate lawsuits from official public platforms in study country (e.g., China Judgments Online; UK BAILII), and measure corporate litigation risk. Our study also provides a fresh perspective on litigation risk in China's capital markets, and expand textual asset pricing research beyond the conventional focus on media texts, corporate disclosures, and policy communications in equity markets. Meanwhile, for future research, it will be of interest to investigate how litigation risk affects time-series variation in aggregate market returns, providing more insights for the role of litigation risk in equity markets.

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Figure 1 An Example of Judgement on China Judgments Online

This figure represents a sample judgment document we obtained from China Judgments Online. Using web crawler technology, we collect text data from 140 million court judgments like this, and extract the plaintiff and defendant information, cause of action, trial procedure, total case amount and other information.

黄小军与重庆永辉超市有限公司两江新区星湖路分公司产品责任纠纷一审民事判决书

案　由 产品责任纠纷

发布日期 2021-01-20

案　号 (2020) 渝0112民初27402号

浏览次数 214

重庆市渝北区人民法院

民 事 判 决 书

（2020）渝0112民初27402号

原告：黄小军，男，汉族，1977年3月7日出生，住四川省宣汉县。

被告：重庆永辉超市有限公司两江新区星湖路分公司，住所地重庆市北部新区高新园星湖路3号附13号，统一社会信用代码915000005842722362。

负责人：周淼。

原告黄小军与被告重庆永辉超市有限公司两江新区星湖路分公司产品责任纠纷一案，本院于2020年10月9日受理后，依法由审判员刘涛峰适用简易程序于2020年12月15日进行公开开庭审理。原告黄小军到庭参加诉讼，被告重庆永辉超市有限公司两江新区星湖路分公司经本院合法传唤无正当理由未到庭参加诉讼，本院依法进行缺席审理。本案现已审理终结。

原告向本院提出诉讼请求：1.判令被告赔偿原告1000元，并返还货款8.30元；2.判令诉讼费由被告承担。事实和理由：原告于2020年8月2日在被告处购买了绿家福新疆灰枣450g一袋，花费8.3元，该商品生产日期为2019年8月1日，保质期12个月，至购买时此商品已超过保质期，此行为违反了《食品安全法》第三十四条、第五十四条的规定，故原告依据我国相关法律规定起诉至法院。

被告重庆永辉超市有限公司两江新区星湖路分公司未作答辩。

当事人围绕其诉讼请求，向本院提交了购物小票、产品实物、购物视频、支付凭证等证据，经庭审审查后，本院认定事实如下：2020年8月2日，原告在被告处购买了一袋绿家福新疆灰枣450g，价格为8.30元，该产品包装上标注：食品名称：新进灰枣，原产地：新疆•喀什，配料：灰枣，规格：450g，执行标准：Q/QD0003S，食品生产许可证编号：SC10350010800546，保质期：见封口或喷码处，产地：重庆市巴南区，委托商：重庆绿家福农业发展有限公司，生产商：重庆琪栋食品有限公司（分装）、贮存条件、食用方法及营养成分表等信息，在该产品封口处标注的生产日期为2019年8月1日。

本院认为，原告于2020年8月2日在被告处购买了绿家福新疆灰枣一袋，双方之间成立买卖合同，该合同合法有效。根据《中华人民共和国食品安全法》第三十四条的规定，禁止生产经营下列食品：……（十）标注虚假生产日期或者超过保质期的食品……。第一百四十八条第二款规定：生产不符合食品安全标准的食品或者经营明知是不符合食品安全标准的食品，消费者除要求赔偿损失外，还可以向生产者或者经营者要求支付价款十倍或者损失三倍的赔偿金；增加赔偿的金额不足一千元的，为一千元……。涉案食品包装上明确标注的产品生产日期为2019年8月1日，保质期为12个月，而原告购买该时（2020年8月2日），该产品已超过保质期，属于不符合食品安全标准的食品。被告作为销售者，明知产品已经超过保质期，仍然销售，应当退还货款，并按照《中华人民共和国食品安全法》第一百四十八条第二款的规定承担1000元的赔偿责任。

据此，依照《中华人民共和国食品安全法》第三十四条、第一百四十八条、《中华人民共和国民事诉讼法》第六十四条、第一百四十四条之规定，判决如下：

被告重庆永辉超市有限公司两江新区星湖路分公司于本判决生效之日起十日内退还原告黄小军货款8.30元，并赔偿原告黄小军1000元。

如果未按本判决指定的期间履行给付义务，应当依照《中华人民共和国民事诉讼法》第二百五十三条之规定，加倍支付迟延履行期间的债务利息。

案件受理费50元，减半交纳25元，由被告重庆永辉超市有限公司两江新区星湖路分公司负担。

如不服本判决，可在判决书送达之日起十五日内，向本院递交上诉状，并按对方当事人的人数提出副本，上诉于重庆市第一中级人民法院。

审 判 员　刘涛峰

二〇二〇年十二月十六日

法官助理　杨　星

书 记 员　简清清

Figure 2 Distribution of Lawsuits Involving Listed Companies during 2010-2021

The calculation of the number of lawsuits includes all instances where listed companies are either defendant or plaintiff.

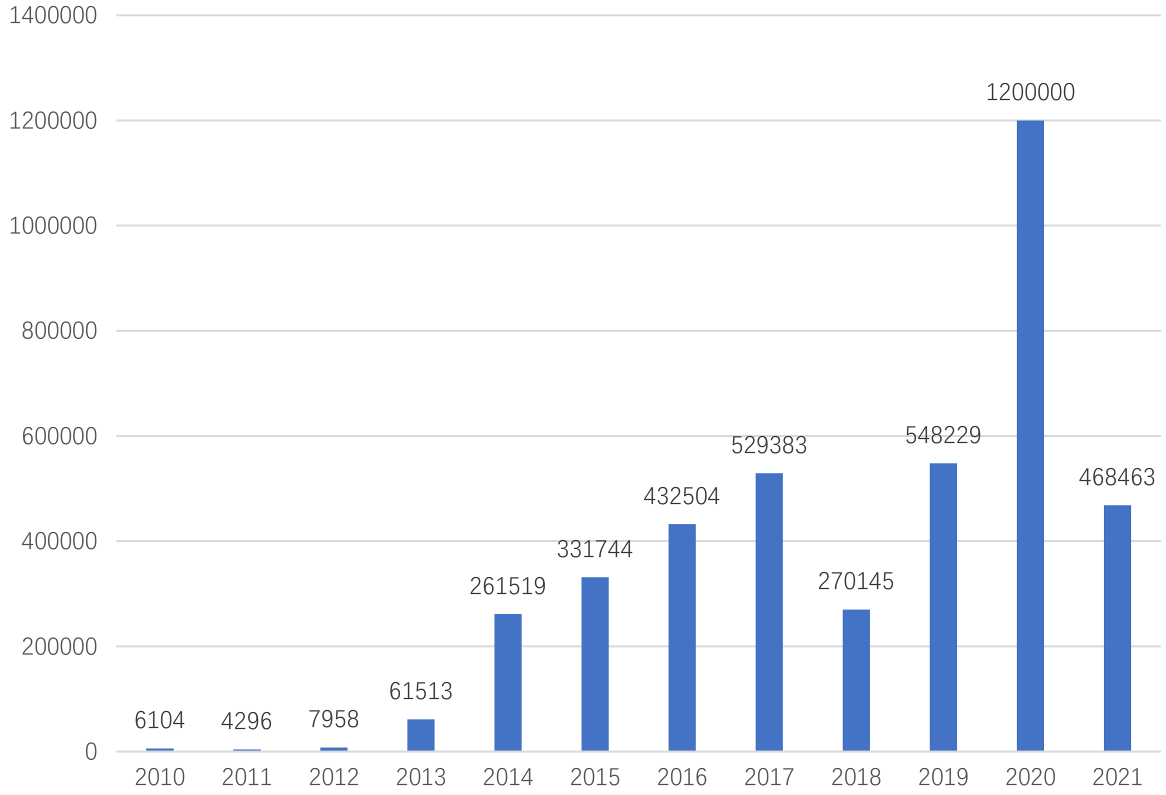

Figure 3 Distribution of Top Twenty Causes of Lawsuits Involving Listed Companies during 2010-2021

This figure depicts the quantitative distribution of litigation causes among Chinese listed companies.

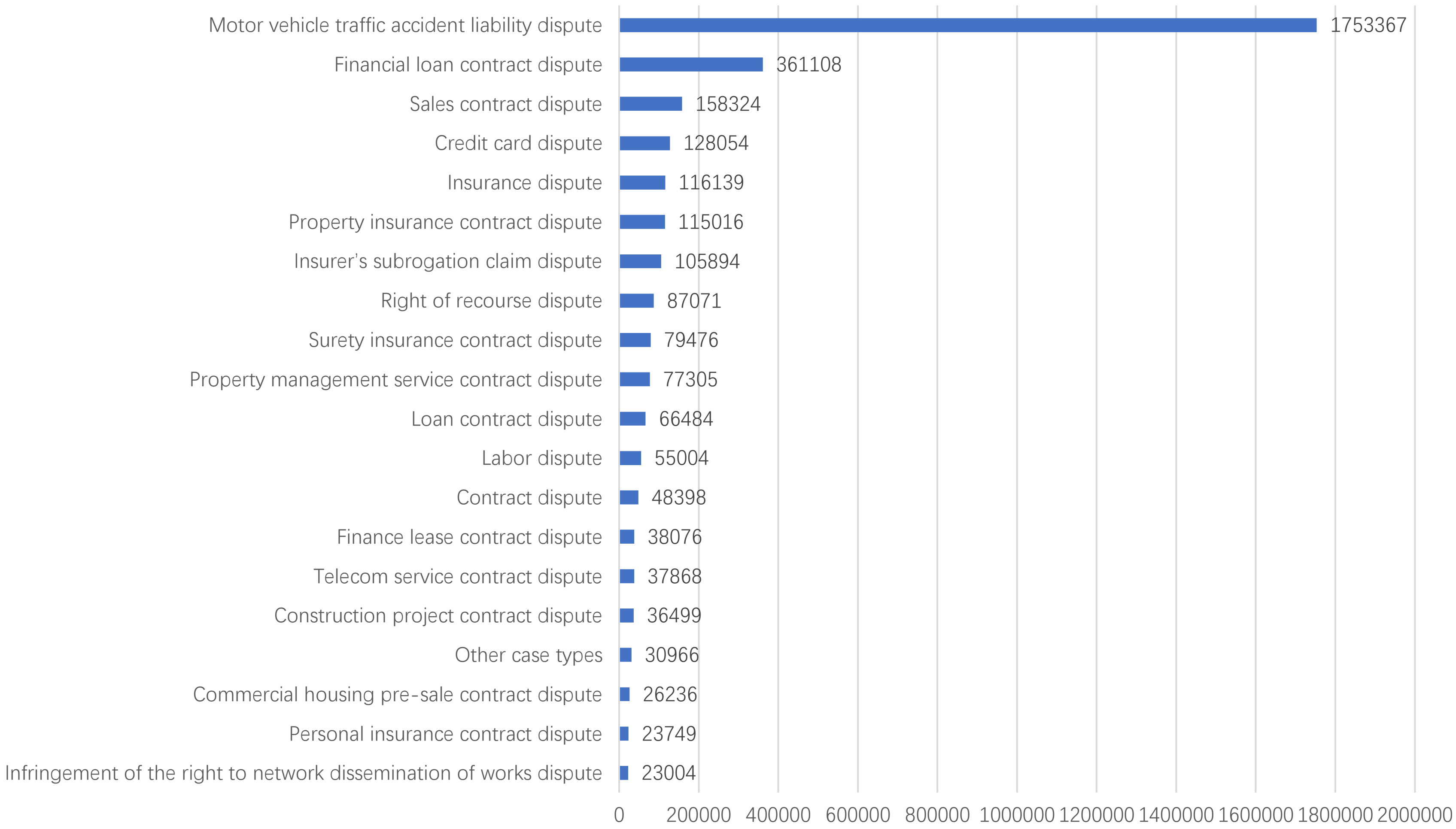

Figure 4 Distribution of Top Twenty Causes of Lawsuits Where Listed Companies Are Defendants during 2010-2021

This figure depicts the quantitative distribution of litigation causes where Chinese listed companies are defendants.

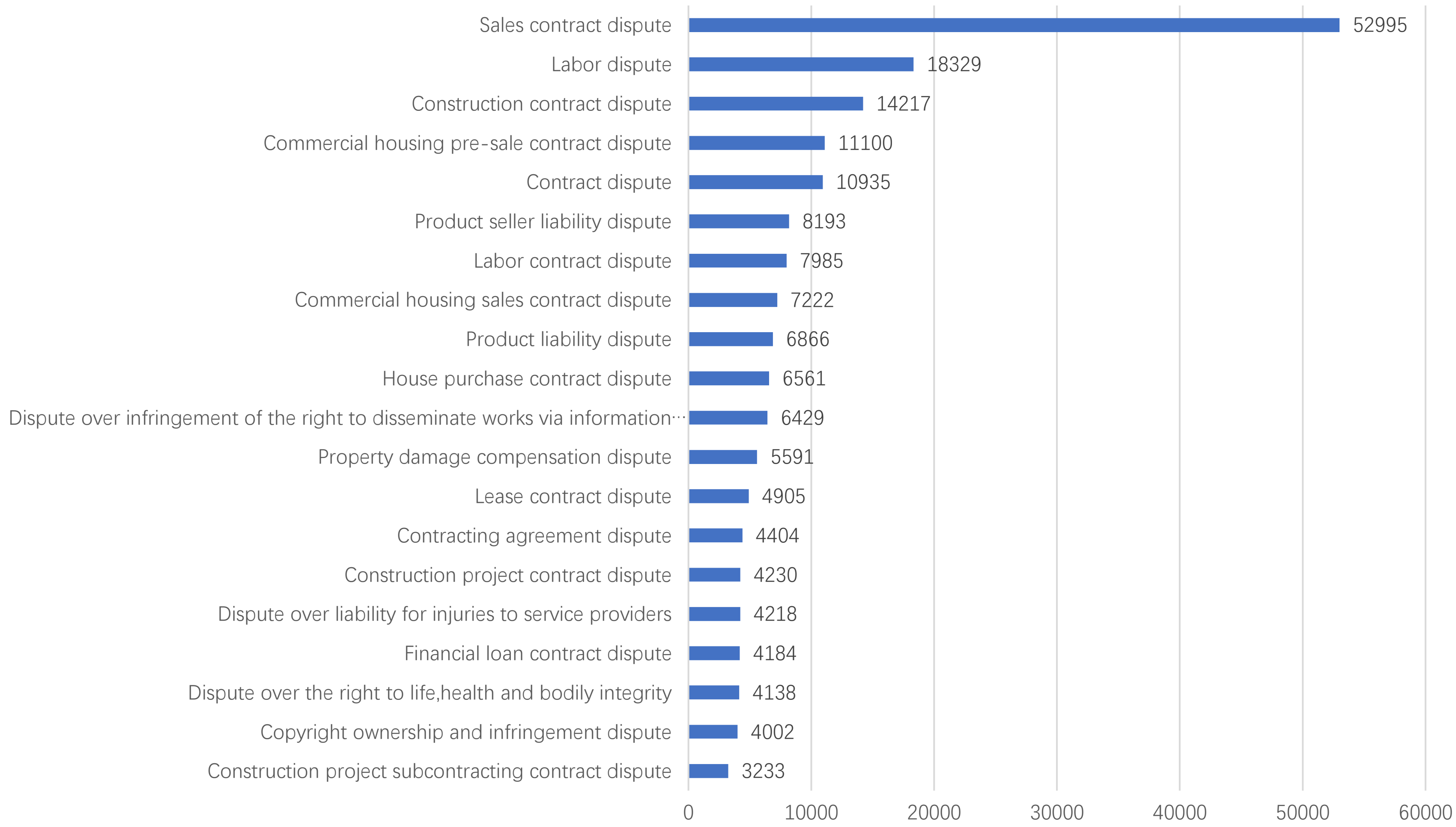

Table 1 Summary Statistics

This table reports summary statistics for the sample used in the analysis including legal and firm variables between 2010 and 2021.

Variables	N	Mean	Std. Dev.	Min	Max
Defendant	2870	0.597	6.401	0	1125
LR	2870	0.408	1.893	0	164.250
Size	2870	22.158	1.442	19.489	27.150
BM	2799	0.454	0.337	0.012	1.834
Age	2755	10.359	7.474	0	27
Growth	2858	0.415	1.202	-0.791	9.007
LEV	2865	0.443	0.220	0.053	0.972
IDR	2865	0.376	0.053	0.333	0.571
SOE	2870	0.361	0.480	0	1
Profitability	2712	0.038	0.029	-0.006	0.152
In (Liabilities)	2809	21.148	1.850	17.382	26.955

Table 2 Univariate Analysis on Litigation Risk by Rebalancing Monthly in the Chinese Stock Market

Monthly returns and alphas are expressed in percentages and t-statistics are in squared brackets. The portfolios' return is calculated by equal weighted. The sample period is 2013:01–2021:08 for CAPM, FF3, FF5 regressions. The "High–low" spread portfolio is computed as the difference between the returns of the highest and lowest litigation risk portfolios.

Portfolios	Return	CAPM α	FF3 α	FF5 α
Low	0.90 [1.14]	-0.21 [-0.64]	-0.59 [-3.58]	-0.52 [-3.30]
Medium	0.59 [0.81]	-0.48 [-1.99]	-0.73 [-4.06]	-0.67 [-3.39]
High	0.57 [0.71]	-0.54 [-1.55]	-0.91 [-4.41]	-0.82 [-4.15]
High-low	-0.33 [-2.22]	-0.33 [-2.27]	-0.32 [-2.19]	-0.31 [-2.04]

Table 3 Bivariate Analysis on Firm Size and Litigation Risk

Monthly returns and alphas are expressed in percentages and t-statistics are in squared brackets. The portfolios' return is calculated by equal weighted. The "High–low" spread portfolio is computed as the difference between the returns of the highest and the lowest litigation risk portfolios within each size group.

	Micro	Small	Large
Panel A：Raw return			
Low	1.77 [1.87]	0.71 [0.85]	0.35 [0.49]
Medium	1.57 [1.64]	0.27 [0.32]	0.61 [0.81]
High	1.25 [1.36]	0.47 [0.55]	0.54 [0.73]
High-low	-0.51 [-2.40]	-0.25 [-1.38]	0.18 [0.94]
Panel B：CAPM α			
Low	0.55 [1.04]	-0.47 [-1.26]	-0.75 [-3.73]
Medium	0.37 [0.65]	-0.93 [-3.13]	-0.49 [-1.58]
High	0.05 [0.10]	-0.71 [-1.80]	-0.58 [-2.58]
High-low	-0.49 [-2.29]	-0.25 [-1.36]	0.17 [0.86]
Panel C：FF3 α			
Low	-0.01 [-0.02]	-0.82 [-3.95]	-0.77 [-3.80]
Medium	-0.19 [-0.51]	-1.19 [-5.23]	-0.46 [-1.91]
High	-0.47 [-1.89]	-1.07 [-4.28]	-0.64 [-2.86]
High-low	-0.47 [-2.19]	-0.25 [-1.38]	0.13 [0.65]
Panel D：FF5 α			
Low	0.08 [0.37]	-0.72 [-3.67]	-0.74 [-3.54]
Medium	-0.02 [-0.06]	-1.10 [-4.59]	-0.48 [-1.83]
High	-0.38 [-1.45]	-0.99 [-4.02]	-0.57 [-2.51]
High-low	-0.47 [-2.09]	-0.27 [-1.44]	0.17 [0.82]

Table 4 Descriptive Statistics of Legal Fee Groups

This table displays the average amount for the low, medium, and high legal fee groups. We break down total legal fee into defendant fee and plaintiff fee, then categorize them into low, medium, and high groups based on the 30th and 70th percentiles for subsequent portfolio analysis.

Group	Type	Average amount (¥)
Low		147.03
Medium	Total legal fee	4018.17
High		2074115.12
Low		5.23
Medium	Defendant fee	1289.80
High		2032956.45
Low		0.70
Medium	Plaintiff fee	414.53
High		46951.56

Table 5 Bivariate Analysis on Legal Fee and Litigation Risk

Monthly returns and alphas are expressed in percentages and t-statistics are in squared brackets. The sample period is 2013:01–2021:08 for CAPM, FF3, FF5 regressions. The portfolios' return is calculated by equal weighted. The "High–low" spread portfolio is computed as the difference between the returns of the highest and the lowest litigation risk portfolios within each legal fee group.

	Low	Medium	High
Panel A： Raw return			
Low	0.94 [1.16]	0.90 [1.08]	1.02 [1.26]
Medium	0.35 [0.50]	0.65 [0.81]	0.59 [0.76]
High	0.91 [1.10]	0.80 [0.97]	0.34 [0.43]
High-low	-0.03 [-1.19]	-0.10 [-0.45]	-0.68 [-2.91]
Panel B： CAPM α			
Low	-0.23 [-0.69]	-0.29 [-0.79]	-0.14 [-0.39]
Medium	-0.66 [-1.90]	-0.49 [-1.32]	-0.57 [-2.02]
High	-0.27 [-0.71]	-0.38 [-1.08]	-0.82 [-2.58]
High-low	-0.03 [-0.18]	-0.10 [-0.43]	-0.68 [-2.86]
Panel C： FF3 α			
Low	-0.56 [-3.20]	-0.63 [-3.27]	-0.46 [-1.94]
Medium	-0.83 [-2.57]	-0.75 [-2.47]	-0.72 [-3.31]
High	-0.59 [-2.61]	-0.70 [-3.03]	-1.09 [-4.75]
High-low	-0.04 [-0.21]	-0.06 [-0.28]	-0.63 [-2.80]
Panel D： FF5 α			
Low	-0.51 [-3.07]	-0.55 [-2.96]	-0.29 [-1.25]
Medium	-0.86 [-2.51]	-0.81 [-2.61]	-0.64 [-2.63]
High	-0.53 [-2.36]	-0.61 [-2.59]	-0.99 [-4.26]
High-low	-0.02 [-1.10]	-0.06 [-0.26]	-0.70 [-3.10]

Table 6 Bivariate Analysis on Defendant Fee and Litigation Risk

Monthly returns and alphas are expressed in percentages and t-statistics are in squared brackets. The sample period is 2013:01–2021:08 for CAPM, FF3, FF5 regressions. The portfolios' return is calculated by equal weighted. The "High–low" spread portfolio is computed as the difference between the returns of the highest and the lowest litigation risk portfolios within each defendant fee group.

	Low	Medium	High
Panel A：Raw return			
Low	1.00 [1.21]	0.78 [0.97]	0.91 [1.11]
Medium	0.65 [0.93]	0.63 [0.80]	0.53 [0.68]
High	0.86 [1.02]	0.86 [1.06]	0.36 [0.45]
High-low	-0.14 [-0.86]	0.08 [0.33]	-0.56 [-2.36]
Panel B：CAPM α			
Low	-0.19 [-0.54]	-0.39 [-1.20]	-0.25 [-0.65]
Medium	-0.36 [-1.00]	-0.50[-1.47]	-0.63 [-2.25]
High	-0.33 [-0.86]	-0.29 [-0.78]	-0.80 [-2.60]
High-low	-0.14 [-0.88]	0.10 [0.42]	-0.56 [-2.33]
Panel C：FF3 α			
Low	-0.51 [-2.90]	-0.67 [-3.26]	-0.61 [-2.70]
Medium	-0.50 [-1.47]	-0.72 [-2.51]	-0.80 [-3.80]
High	-0.68 [-3.04]	-0.57 [-2.07]	-1.06 [-4.60]
High-low	-0.16 [-1.00]	0.10 [0.42]	-0.45 [-2.16]
Panel D：FF5 α			
Low	-0.46 [-2.75]	-0.63 [-3.05]	-0.43 [-1.93]
Medium	-0.49 [-1.39]	-0.68 [-2.24]	-0.75 [-3.14]
High	-0.62 [-2.86]	-0.47 [-1.69]	-0.96 [-3.98]
High-low	-0.16 [-0.97]	0.16 [0.68]	-0.53 [-2.50]

Table 7 Bivariate Analysis on Win Rate and Litigation Risk

Monthly returns and alphas are expressed in percentages and t-statistics are in squared brackets. The sample period is 2013:01–2021:08 for CAPM, FF3, FF5 regressions. The portfolios' return is calculated by equal weighted. The "High–low" spread portfolio is computed as the difference between the returns of the highest and the lowest litigation risk portfolios within each win rate group.

	Rate<=50%	Rate>50%
Panel A： Raw return		
Low	0.88 [1.02]	0.88 [1.01]
Medium	0.44 [0.56]	1.04 [1.28]
High	0.56 [0.66]	0.64 [0.73]
High-low	-0.32 [-2.15]	-0.24 [-1.29]
Panel B： CAPM α		
Low	-0.37 [-0.99]	-0.38 [-1.06]
Medium	-0.76 [-2.75]	-0.14[-0.37]
High	-0.68 [-1.92]	-0.63 [-1.68]
High-low	-0.31 [-2.06]	-0.24 [-1.30]
Panel C： FF3 α		
Low	-0.51 [-2.87]	-0.51 [-2.35]
Medium	-0.81 [-3.83]	-0.18 [-0.56]
High	-0.80 [-3.88]	-0.75 [-3.39]
High-low	-0.29 [-2.04]	-0.25 [-1.31]
Panel D： FF5 α		
Low	-0.41 [-2.48]	-0.38 [-1.86]
Medium	-0.76 [-3.30]	-0.19 [-0.51]
High	-0.70 [-3.44]	-0.70 [-3.14]
High-low	-0.28 [-1.96]	-0.31 [-1.61]

Table 8 Regression of Company Profitability and Litigation Risk

This table reports results from following regression,

$$Profitability_{i,t} = \beta_0 + \beta_1 LR_{i,t-1} + \beta_2 Controls_{i,t-1} + \mu_t + \gamma_i + \epsilon_{i,t} ,$$

where $Profitability_{i,t}$ refers to the profitability of company i in month t, $LR_{i,t-1}$ refers to the litigation risk of company i in month $t-1$ which is measured by equation (1), $Controls_{i,t-1}$ refers to the control variables listed in Table 1, μ_t refers to the time fixed effect, γ_i refers to the entity fixed effect. The slopes of variable are expressed in percentages and t-statistics are in squared brackets. The sample period is from January 2010 to December 2021.

	Profitability			
VARIABLES	(1)	(2)	(3)	(4)
$LR_{i,t-1}$	-0.004***	-0.004***		
	(-2.58)	(-2.86)		
$Defendant_{i,t-1}$			-0.005***	-0.001*
			(-9.36)	(-1.71)
Controls	No	Yes	No	Yes
Time Effects	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes
Observations	390,579	383,328	390,579	383,328

Table 9 Regression of Company Total Liabilities and Litigation Risk

This table reports results from following regression,

$$In(Liabilities)_{i,t} = \beta_0 + \beta_1 LR_{i,t-1} + \beta_2 Controls_{i,t-1} + \mu_t + \gamma_i + \epsilon_{i,t} ,$$

where $In(Liabilities)_{i,t}$ refers to the natural logarithm of total liabilities of company i in month t, $LR_{i,t-1}$ refers to the litigation risk of company i in month $t-1$ which is measured by equation (1), $Controls_{i,t-1}$ refers to the control variables listed in Table 1, μ_t refers to the time fixed effect, γ_i refers to the entity fixed effect. The slopes of variable are expressed in percentages and t-statistics are in squared brackets. The sample period is from January 2010 to December 2021.

	Total liabilities			
VARIABLES	(1)	(2)	(3)	(4)
$LR_{i,t-1}$	0.413***	0.393***		
	(7.81)	(7.50)		
$Defendant_{i,t-1}$			0.133***	0.111***
			(7.35)	(6.18)
Controls	No	Yes	No	Yes
Time Effects	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes
Observations	404,494	396,458	404,485	396,458

Appendix:

Table A1 Univariate Analysis on Litigation Risk by Rebalancing Monthly in the Chinese Stock Market

Monthly returns and alphas are expressed in percentages and t-statistics are in squared brackets. The sample period is 2013:01–2021:08 for CAPM, FF3, FF5 regressions by value weighted. The "High–low" spread portfolio is computed as the difference between the returns of the highest and lowest litigation risk portfolios.

Portfolios	Return	CAPM α	FF3 α	FF5 α
Low	0.40 [0.58]	-0.65 [-3.49]	-0.72 [-4.09]	-0.67 [-3.80]
Medium	0.51 [0.73]	-0.53 [-2.13]	-0.44 [-2.29]	-0.49 [-2.33]
High	0.38 [0.51]	-0.71 [-2.57]	-0.77 [-3.10]	-0.71 [-2.97]
High-low	-0.02 [-0.08]	-0.06 [-0.26]	-0.06 [-0.27]	-0.04 [-0.17]

Table A2 Bivariate Analysis on Plaintiff Fee and Litigation Risk

Monthly returns and alphas are expressed in percentages and t-statistics are in squared brackets. The sample period is 2013:01–2021:08 for CAPM, FF3, FF5 regressions. The portfolios' return is calculated by equal weighted. The "High–low" spread portfolio is computed as the difference between the returns of the highest and the lowest litigation risk portfolios within each plaintiff fee group.

	Low	Medium	High
Panel A：Raw return			
Low	0.94 [1.15]	1.06 [1.27]	0.91 [1.11]
Medium	1.01 [1.28]	-0.05 [-0.05]	0.61 [0.80]
High	0.80 [0.98]	0.90 [1.05]	0.55 [0.69]
High-low	-0.14 [-0.97]	-0.16 [-0.49]	-0.36 [-1.55]
Panel B：CAPM α			
Low	-0.24 [-0.68]	-0.11 [-0.29]	-0.27 [-0.81]
Medium	-0.10 [-0.24]	-1.16[-1.61]	-0.54 [-2.03]
High	-0.37 [-1.03]	-0.31 [-0.82]	-0.60 [-1.73]
High-low	-0.13 [-0.91]	-0.20 [-0.61]	-0.32 [-1.37]
Panel C：FF3 α			
Low	-0.57 [-3.25]	-0.43 [-1.57]	-0.59 [-3.06]
Medium	-0.41 [-1.25]	-1.32 [-1.82]	-0.67 [-3.05]
High	-0.69 [-3.13]	-0.62 [-2.51]	-0.89 [-3.52]
High-low	-0.12 [-0.83]	-0.20 [-0.60]	-0.30 [-1.33]
Panel D：FF5 α			
Low	-0.49 [-2.94]	-0.41 [-1.46]	-0.49 [-2.69]
Medium	-0.24 [-0.69]	-1.43 [-1.94]	-0.65 [-2.63]
High	-0.59 [-2.77]	-0.63 [-2.53]	-0.78 [-2.95]
High-low	-0.10 [-0.64]	-0.22 [-0.67]	-0.29 [-1.23]

Table A3 Regression of Company Profitability and Litigation Risk over Different Time Intervals

This table reports results from following regression,

$$Profitability_{i,t} = \beta_0 + \beta_1 LR_{i,t-h} + \beta_2 Controls_{i,t-h} + \mu_t + \gamma_i + \epsilon_{i,t} ,$$

where $Profitability_{i,t}$ refers to the profitability of company i in month t, $LR_{i,t-h}$ refers to the litigation risk of company i in month $t-h$ which is measured by equation (1), $h = 6, 12$, $Controls_{i,t-h}$ refers to the control variables listed in Table 1, μ_t refers to the time fixed effect, γ_i refers to the entity fixed effect. The slopes of variable are expressed in percentages and t-statistics are in squared brackets. The sample period is from January 2010 to December 2021.

	Profitability			
VARIABLES	(1)	(2)	(3)	(4)
$LR_{i,t-6}$	-0.004**			
	(-2.49)			
$LR_{i,t-12}$		-0.007***		
		(-4.30)		
$Defendant_{i,t-6}$			-0.001*	
			(-1.87)	
$Defendant_{i,t-12}$				-0.001
				(-1.06)
Controls	Yes	Yes	Yes	Yes
Time Effects	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes
Observations	383,328	383,328	383,328	383,328

Table A4 Regression of Company Total Liabilities and Litigation Risk over Different Time Intervals

This table reports results from following regression,

$$In(Liabilities)_{i,t} = \beta_0 + \beta_1 LR_{i,t-h} + \beta_2 Controls_{i,t-h} + \mu_t + \gamma_i + \epsilon_{i,t} ,$$

where $In(Liabilities)_{i,t}$ refers to the natural logarithm of total liabilities of company i in month t, $LR_{i,t-h}$ refers to the litigation risk of company i in month $t-h$ which is measured by equation (1), $h = 6, 12$, $Controls_{i,t-h}$ refers to the control variables listed in Table 1, μ_t refers to the time fixed effect, γ_i refers to the entity fixed effect. The slopes of variable are expressed in percentages and t-statistics are in squared brackets. The sample period is from January 2010 to December 2021.

	Total liabilities			
VARIABLES	(1)	(2)	(3)	(4)
$LR_{i,t-6}$	0.395***			
	(7.41)			
$LR_{i,t-12}$		0.402***		
		(7.27)		
$Defendant_{i,t-6}$			0.084***	
			(4.66)	
$Defendant_{i,t-12}$				0.071***
				(3.78)
Controls	Yes	Yes	Yes	Yes
Time Effects	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes
Observations	396,458	396,458	396,458	396,459